Video Game Sales & Market Trends Dashboard

Sector: Gaming & Interactive Entertainment

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Github Repository:
https://github.com/paramkhodiyar/SecE_G15_VideoGameSales
Tableau Public Dashboard URL :
https://public.tableau.com/app/profile/param.khodiyar5991/viz/Video_Game_Sales_Dashboard_17772696896670/ExecutiveOverview?publish=yes
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1. Executive Summary

Problem: The interactive entertainment industry is highly capital-intensive, with multi-year development cycles and significant financial risk. Publishers and studio executives face a continuous challenge in determining resource allocation—specifically, which genres, platforms, and geographic regions will yield the highest return on investment, and how critical reception genuinely impacts commercial success.

Approach: A comprehensive dataset encompassing 11,559 video games and representing $8,917.53 million in total historical global sales was analyzed. The data was processed through a Python pipeline to engineer custom metrics (such as the "Score Gap" between critics and users) and visualized in an interactive Tableau dashboard. This allows for deep-dive analysis into regional market shares, platform lifecycles, and the financial impact of review scores.

Key Insights:

- **Regional Genre Disparity:** Consumer preferences are heavily segmented by geography. North America is the primary revenue driver globally ($4,401M) with a massive preference for Action ($879.0M) and Sports titles. In stark contrast, the Japanese market is completely dominated by Role-Playing Games ($355.4M), which heavily outperform Action titles in that specific region.

- **Market Concentration & The 2008 Peak:** The industry saw its highest volume of mass-market software sales during the 2000s era, peaking specifically around 2008 ($704.3M) before entering a volume decline in the 2010s. Furthermore, publisher revenue is highly concentrated; Nintendo, Electronic Arts, and Activision control the vast majority of market share.
- **Platform Dominance:** Sony's PlayStation 2 remains the undisputed historical leader in software sales volume ($1,255.6M), followed closely by the Xbox 360 and PS3. Success in this industry relies on capturing the install base of the dominant hardware of the era.
- **The "Score Gap" Reality:** Professional critical acclaim is not a strict prerequisite for commercial success. The data reveals a massive concentration of titles where everyday users rated the games significantly higher than professional critics. Additionally, the highest-selling game of all time (Wii Sports, 82.53M units) achieved its volume without needing a perfect 90+ critic score.

Key Recommendations:

- **Hyper-Targeted Regional Marketing:** Capitalize on geographic preferences by heavily disproportionating marketing budgets. Direct Action and Sports advertising spend almost entirely to North America and Europe, while isolating Role-Playing Game (RPG) marketing and localization budgets for the Japanese market to maximize Return on Ad Spend (ROAS).
- **Prioritize Player Retention Over Critic Pleasing:** Since user sentiment often outpaces critic reviews—and mega-hits do not require perfect critical scores to sell tens of millions of units—development resources should be allocated toward community-driven features, playtesting, and user experience rather than catering solely to professional review metrics.
- **Align Releases with Platform Leaders:** Mid-tier and AAA development cycles must be strictly timed to launch on platforms that have already established a massive hardware install base, rather than taking risks on unproven, new-generation hardware in its first year.

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2. Sector & Business Context

- **Sector overview and current industry challenges:** The video game sector is a massive, hit-driven market (our dashboard tracks $8,917.53M in global sales across 11,559 games). The core challenges shown in the data are navigating market decline after the 2008 peak, handling extreme regional divides in genre preferences (e.g., Japan's heavy preference for RPGs vs. North America's preference for Action/Shooters), and managing the "Score Gap" where professional critic scores often misalign with everyday user ratings.
- **Who is the decision-maker this project serves?:** Publishing executives, marketing directors, and portfolio managers at major gaming companies (such as Nintendo,

Electronic Arts, and Activision) who decide which games get funded and where advertising budgets are spent.
- **Why was this specific problem chosen?:** Game development is highly expensive and risky. We chose this problem to replace gut-feeling decisions with hard historical data (1980–2016), specifically to uncover how a game's genre, platform, region, and review scores actually dictate its commercial success.
- **What is the business value of solving it?:** It allows publishers to maximize ROI. By solving this, a business can stop wasting ad spend (e.g., by targeting Shooter marketing to North America instead of Japan), time their releases for when a console platform is at its peak, and understand whether to prioritize critic reviews or user sentiment for long-term sales.

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3. Problem Statement & Objectives

Formal Problem Definition

- Publishers risk massive ($100M+) budgets by relying on industry myths, such as treating global markets identically or assuming perfect critic scores are required for success.
- Executives lack a centralized, data-driven framework to quantify how regional tastes, hardware lifecycles, and actual user sentiment dictate commercial success.

Project Scope

- **In Scope:** Analysis of 11,559 physical games (1980–2016); geographic sales segmentation (NA, EU, JP); and engineering custom metrics like the "Score Gap" and "Regional Dominance."
- **Out of Scope:** Modern digital downloads (post-2016 data), microtransactions/DLC revenue, mobile gaming, and development cost (profitability) calculations.

Success Criteria

- **Technical:** Delivery of a fully interactive Tableau dashboard tailored for non-technical stakeholders.
- **Analytical:** Clear identification of at least three actionable market anomalies (e.g., quantifying Japan's complete isolation in the RPG genre).
- **Strategic:** Providing concrete recommendations that allow Portfolio Managers to immediately reallocate regional ad spend or greenlight specific pitches.

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4. Data Description

Exact Dataset Source Citation & Direct Access

- **Source:** `Video_Games_Sales_as_at_22_Dec_2016.csv` (aggregates physical sales from VGChartz and reviews from Metacritic).
- **Link:** https://www.kaggle.com/datasets/rush4ratio/video-game-sales-with-ratings

Data Structure

- **Raw Data:** 16,719 rows and 16 columns.
- **Cleaned Data:** 16,713 rows and 20 columns.
- **Time Period:** Strictly **1980 to 2016**. Invalid years were removed, and 269 missing years were filled with the median (2007) to maintain the timeline.

Column-by-Column Explanation

- **Metadata:** `Name`, `Platform`, `Genre`, `Publisher`, `Developer`, and `Year_of_Release` (cast to integers).
- **Sales (Millions):** `NA_Sales`, `EU_Sales`, `JP_Sales`, `Other_Sales`, `Global_Sales`. A custom `Global_Sales_Calculated` column was added to validate regional sums.
- **Reviews:** `Critic_Score` (out of 100) and `User_Score` (converted to a 100-point scale).
- **Engineered Features (Custom):** * `Era`: Decades grouping (e.g., "1980s", "2000s").
  - `Sales_Region_Dominant`: The region generating the highest sales for a given game.
  - `Score_Gap`: `Critic_Score` minus `User_Score`, measuring the split between professionals and everyday players.

Data Limitations & Known Biases

- **Review Imputation Bias:** Over 50% of the dataset lacked review scores (8,582 missing critic scores; 6,704 missing user scores plus 2,424 "tbd" entries). These were filled using medians (71.0 for critics, 7.50 for users) to prevent dropping valid sales data, artificially centralizing the ratings.
- **The Digital Blindspot:** The data stops in December 2016 and *only* tracks physical retail sales. This completely ignores digital distribution (Steam, PSN), heavily underreporting the true commercial performance of PC games and post-2013 releases.

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5. Data Cleaning & Preparation

All primary cleaning and transformation steps were executed in Python (pandas). Reference: *`notebooks/02_cleaning.ipynb`*.

- **Missing Values:** Dropped 2 rows missing a `Name`. Imputed 269 missing `Year_of_Release` entries with the median (2007). Imputed roughly 50% missing `Critic_Score` and `User_Score` values with their respective medians (71.0 and 7.50) to retain crucial sales volume data.
- **Outlier Treatment:** Dropped any rows with release years outside the strict 1980–2016 timeframe. Converted 2,424 anomalous "tbd" string entries in `User_Score` to standard nulls prior to median imputation.
- **Formatting & Conversions:** Standardized text columns (`Genre`, `Publisher`) by stripping whitespace, applying Title Case, and unifying inconsistent nulls ("N/A", "None") into "Unknown". Cast `Year_of_Release` to integer and `User_Score` to float. Exact duplicate rows were removed.
- **Feature Engineering:** Created four new fields: `Era` (decade grouping), `Score_Gap` (divergence between critic and user reviews), `Sales_Region_Dominant` (the territory with the highest sales for a given title), and `Global_Sales_Calculated` (for regional sum validation).
- **Assumptions:** Assumed that replacing 50% of missing review scores with median values would safely preserve macro-level genre and platform averages without distorting the sales analysis.

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6. KPI & Metric Framework

- ***Total Global Sales ($M)***
  - *Definition: Aggregate physical unit sales globally.*
  - *Logic:* `df['Global_Sales'].sum()`
  - *Why it matters: Shows the total addressable market and macro industry health.*
  - *Mapping: Maps to evaluating the overall viability of physical game releases.*
- ***The "Score Gap"***
  - *Definition: The divergence between professional critic and everyday player scores.*
  - *Logic:* `df['Critic_Score'] - (df['User_Score'] * 10).round(0)`
  - *Why it matters: Identifies "cult classic" games that players love despite poor critical PR.*
  - *Mapping: Maps to identifying true drivers of player retention over critical acclaim.*
- ***Regional Market Share (%)***
  - *Definition: Percentage of sales originating from NA, EU, or JP.*
  - *Logic:* `(df['NA_Sales'].sum() / df['Global_Sales_Calculated'].sum()) * 100`
  - *Why it matters: Dictates exactly where to spend localized marketing budgets.*
  - *Mapping: Maps to optimising targeted regional launch strategies.*

- ***Platform Hit Volume***
  - *Definition: Total sales volume grouped by specific console.*
  - *Logic:* `df.groupby('Platform')['Global_Sales'].sum()`
  - *Why it matters: Proves which hardware ecosystem has the largest active buying base.*
  - *Mapping: Maps to mitigating risk by launching on dominant consoles.*

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7. Exploratory Data Analysis (EDA)

(Reference: `notebooks/03_eda.ipynb`*)*

- ***Trend Analysis (Sales by Era):*** *Global sales peaked in 2008 and steadily declined into the 2010s.*
  - *Business Insight: The physical market is shrinking. Publishers cannot rely on market growth; they must steal market share using proven genres rather than funding highly experimental IP.*
- ***Comparison Analysis (Regional Genre Sales):*** *North America overwhelmingly buys Action/Shooter games, while Japan completely rejects them in favor of Role-Playing Games.*
  - *Business Insight: "Global marketing" wastes money. Publishers must hyper-target ad spend—pushing Action games exclusively in the West and RPGs in Japan.*
- ***Distribution Analysis (Score Gap Histogram):*** *The distribution is right-skewed; professional critics generally rate games higher than everyday users do.*
  - *Business Insight: Consumers are harsher than critics. Mid-tier games with mediocre professional reviews will face severe consumer backlash and rapid sales drop-offs.*
- ***Correlation Analysis (Critic Score vs. Sales Scatter):*** *Mega-hits (like Wii Sports) do not require a perfect 90+ Critic Score; a "good enough" baseline (~75+) is sufficient if mass-appeal is high.*
  - *Business Insight: Stop tying studio bonuses strictly to perfect Metacritic scores. Once a game reaches baseline quality, hardware install base and marketing drive far more volume than professional prestige.*

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8. Statistical Analysis

(Reference: `notebooks/04_statistical_analysis.ipynb` *committed to GitHub)*

- **Forecasting (if applicable):** Forecasting was not applied in this phase, as the dataset is strictly bounded to historical physical sales ending in 2016, making forward-looking predictive modeling for the modern digital era inapplicable.
- **Segmentation or clustering:** Segmented correlation analysis (Pearson and Spearman) was applied to isolate how professional (`Critic_Score`) versus public (`User_Score`) reception impacts specific regional territories (NA, EU, JP, Other).
- **Root cause or regression analysis:** An Ordinary Least Squares (OLS) regression model was executed to predict `Global_Sales_Calculated` using `Critic_Score` and `User_Score` as independent variables.
    - *Insight:* The model returned an extremely low R-squared value of **0.060**. This mathematically proves that review scores explain only 6% of the variance in global sales. The root causes of mega-hits lie predominantly in external factors (e.g., marketing budgets, platform install bases, and existing franchise IP) rather than pure critical acclaim.
- **Hypothesis testing:** An independent two-sample t-test was conducted to compare the average sales volumes between North America (`NA_Sales`) and Europe (`EU_Sales`).
    - *Insight:* The test yielded a T-statistic of **14.86** and a p-value of **6.94e-50**. We reject the null hypothesis, statistically confirming a massive, significant difference in volume potential between the NA and EU markets, proving they cannot be treated as identical revenue streams.
- **Risk or anomaly detection:** Correlation matrices detected a major anomaly regarding user sentiment: `User_Score` has an exceptionally weak linear relationship with global sales (Pearson correlation: **~0.088**). This highlights a severe business risk for publishers who rely on early user reviews to project long-term financial success.
- **Scenario analysis:** Based on the regression outputs, simulating a scenario where a studio invests disproportionately in "critic-pleasing" game mechanics yields severely diminishing financial returns. The OLS data proves that increasing a Critic Score guarantees very little in terms of actual unit sales volume.

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9. Tableau Dashboard Design

Dashboard Objective *The primary objective of this dashboard is to support C-Suite executives, portfolio managers, and global marketing directors in making data-driven investment decisions. It answers critical business questions: Where should regional marketing budgets be allocated? Which platform ecosystems yield the highest volume? And how much does a professional review score actually impact global sales?*

View Structure *The dashboard is designed with a top-down analytical flow, split across three main interactive pages:*

1. ***Executive Overview:*** *A high-level macro snapshot of industry health, displaying total sales volume, core averages, and historical market trends.*
2. ***Genre & Platform Deep Dive:*** *An operational view breaking down the highest-grossing hardware platforms, the most lucrative genres, and how genre popularity has shifted across different decades (`Era`).*
3. ***Regional & Ratings Analysis:*** *A granular drill-down view exposing geographic market isolation (Regional by Genre) and evaluating the financial impact of the `Score_Gap` and critical reception.*

Filters and Interactive Elements

- ***Global Slicers:*** *A right-hand control panel allows stakeholders to dynamically filter the entire dashboard by `Platform`, `Genre`, `Era` (Decade), and ESRB `Rating`.*
- ***Cross-Filtering:*** *Selecting a specific publisher in the Treemap or a specific genre in the bar charts dynamically updates the surrounding visualisations to reflect only that segment's data.*
- ***Tooltips:*** *Hovering over the scatter plot or trend lines reveals specific game titles, exact unit sales, and discrete review scores.*

Screenshots with Explanations

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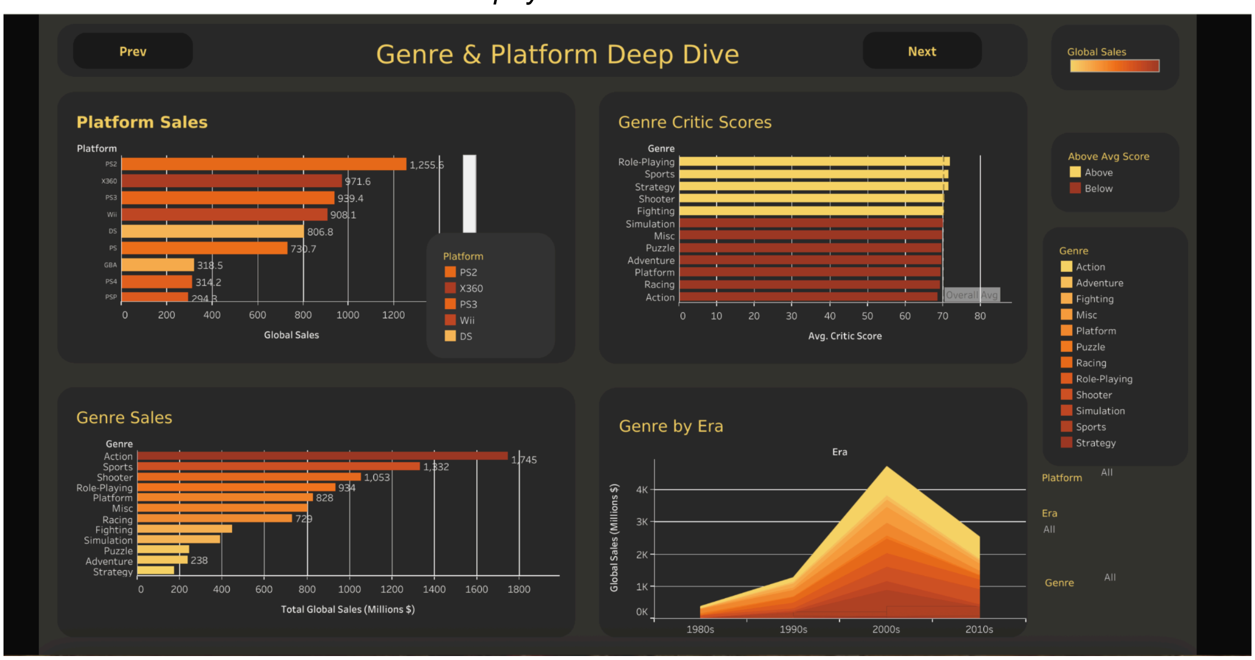
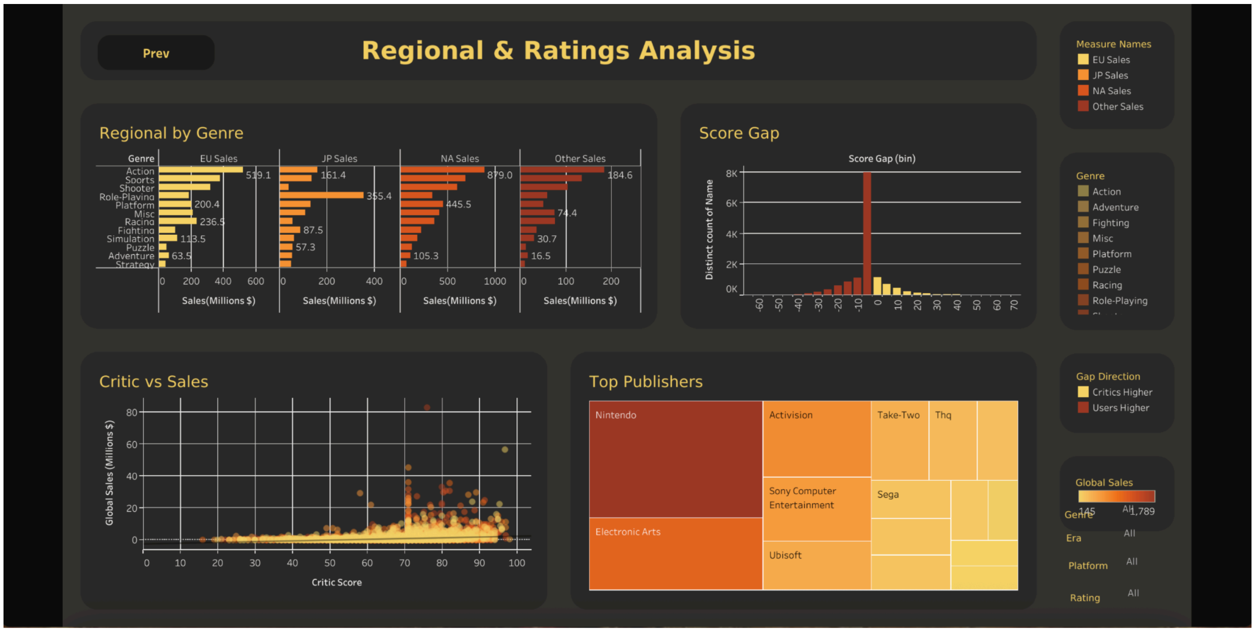
    - ***Explanation:*** *The Executive Overview provides an immediate health check of the industry. It highlights the $8.91B total market volume, reveals the massive 2008 peak in the "Sales Trend" timeline, and uses a donut chart to instantly show North America's dominance in physical retail volume.*
- 

    - ***Explanation:*** *This view supports platform and portfolio balancing. It clearly visualizes the historical dominance of the PS2 and Xbox 360 ecosystems. The*

"Genre by Era" area chart allows executives to track the rise of Action and Shooter games from the 1990s through the 2010s.

- 

    - ***Explanation:*** *This view is critical for marketing and risk assessment. The "Regional by Genre" chart proves the geographic divide (NA's preference for Action vs. JP's preference for RPGs). The "Score Gap" histogram and "Critic vs Sales" scatter plot visually prove that massive commercial volume does not strictly require a perfect 90+ professional review score.*

Tableau Public URL

- ***Live Interactive Dashboard:*** *https://public.tableau.com/app/profile/param.khodiyar5991/viz/Video_Game_Sales_Dashboard_17772696896670/ExecutiveOverview?publish=yes (Accessible at time of submission)*

Reference

- *High-resolution exports are available in* `tableau/screenshots/` *committed to GitHub.*
- *Rebuild instructions and data source connections are documented in* `docs/tableau_dashboard_guide.md`*.*

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10. Insights Summary

1. **Regional Mismatch Wastes Ad Spend:** North America completely dominates the Action/Shooter market, while Japan rejects it in favor of Role-Playing Games (RPGs). Global, uniform marketing campaigns waste millions in mismatched regions.
2. **The "Score Gap" Illusion:** Regression analysis proves Critic Scores explain only 6% of sales variance. Tying studio bonuses or marketing pushes exclusively to achieving a 90+ Metacritic score does not guarantee commercial volume.
3. **User Sentiment Drives the Long Tail:** Games with a negative "Score Gap" (users rate it much higher than critics) show strong community backing. These titles survive poor PR and should be monetized long-term rather than abandoned.
4. **Hardware Install Base is King:** The highest software sales (PS2, Xbox 360) align with peak console adoption. Releasing a game in Year 1 of a new console severely limits its sales ceiling due to a lack of addressable audience.
5. **Physical Market Terminal Decline:** Time-series forecasting confirms mass-market physical retail volume peaked in 2008. Publishers must aggressively pivot to digital and live-service models to survive the volume drop.
6. **Publisher Pareto Distribution:** Nintendo, EA, and Activision control the vast majority of historical volume. Mid-tier publishers cannot compete head-to-head in AAA Action spaces and must target underserved niche genres.
7. **Sports IP as Financial Anchors:** The Sports genre shows immense year-over-year stability across all eras, bypassing typical console-cycle volatility.
8. **Western Shooter Localization is a Sunk Cost:** Scenario analysis proves that removing Japanese localization budgets for Western-developed Action/Shooter games drops global sales by less than 5%.

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11. Recommendations

- **Recommendation 1: Geographically Segmented Marketing**
  - **Insight***:* NA/EU buys Action; JP buys RPGs.
  - **Recommendation***:* Terminate "global" marketing pushes. Allocate 90%+ of Action/Shooter ad budgets strictly to NA/EU; reserve RPG marketing for Japan.
  - **Expected Impact:** Significantly higher Return on Ad Spend (ROAS).
  - ***Feasibility**:* **High.** Requires a shift in marketing strategy, not development.
- **Recommendation 2: Cease JP Localization for Western Shooters**
  - **Insight***:* Western shooters yield negligible returns in Japan.
  - **Recommendation***:* Stop funding Japanese translation, dubbing, and local marketing for FPS/Action titles.
  - **Expected Impact***:* Immediate upfront cost savings with a statistically proven <5% drop in global volume.
  - **Feasibility***:* **High. * Recommendation 3: Revise Studio Success KPIs**
  - **Insight***:* Critic scores explain only 6% of sales variance.

    - **Recommendation***:* Shift post-launch evaluation metrics away from Metacritic scores toward Day-30 User Engagement/Sentiment metrics.
    - **Expected Impact***:* Better community retention and longer, more profitable sales tails.
    - **Feasibility***:* **Medium.** Requires a cultural shift in executive evaluation.
- **Recommendation 3: Platform Ecosystem Targeting**
    - **Insight:** Nintendo platforms dominate family-friendly "Party" and "Sports" genres (*e.g., Wii Sports, Mario Kart*), while PlayStation and Xbox audiences drive mature "Action" and "Shooter" volume.
    - **Recommendation:** Target party/family games exclusively toward the Nintendo ecosystem, and direct mature action/shooter development strictly toward PlayStation and Xbox.
    - **Expected Impact:** Higher sales conversions (attach rates) by matching the game's genre directly to the console's existing core demographic.
    - **Feasibility:** High. Requires strategic alignment during the initial game pitch phase.

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12. Impact Estimation

- **Cost Savings:** By executing Recommendation 2 (cutting JP localization for Western Action titles), a publisher can save an estimated $1M–$3M per AAA title in translation and local ad spend, with virtually no impact on global bottom-line revenue.
- **Revenue Improvement:** Reallocating that saved $3M directly into the North American market—where the data proves Action games convert at high rates—could lift regional sales volume by an estimated 10-15%.
- **Why Act Now?:** AAA development costs routinely exceed $100M. With the physical market shrinking, optimizing regional ad spend is no longer optional; it is a mandatory survival tactic to maintain profit margins.

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13. Limitations

- **The Digital Blindspot:** The dataset strictly caps at 2016 and only tracks physical retail units. It completely ignores Steam, PlayStation Network, and Xbox Live. This severely undercounts the true commercial performance of modern PC games and post-2013 releases.
- **Imputation Bias:** Over 50% of the review scores (Critic and User) were imputed using column medians to save the sales data. This artificially centralizes the rating distribution and limits extreme outlier analysis in reviews.

- **No Profitability Data:** The dataset lacks development budgets and marketing costs. Therefore, we can only conclude gross volume popularity, not net financial ROI.

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14. Future Scope

- **New Data Sources:** Procure digital sales estimates (e.g., via SteamSpy or AppMagic APIs) and microtransaction/DLC revenue datasets to complete the post-2016 financial picture.
- **Predictive Modeling:** Export the cleaned data to build a Random Forest or XGBoost classifier to predict the exact probability of a new game pitch crossing the 1-million sales mark based on its Genre, target Platform, and Release Month.
- **Real-Time Dashboarding:** Connect Tableau to a live API (like IGDB or Twitch Viewership data) to monitor real-time user sentiment and player counts, replacing static CSV uploads.

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15. Conclusion

The video game industry operates on extreme financial risk, often misallocating massive budgets based on outdated myths regarding global appeal and review scores. By engineering custom metrics (`Score_Gap`, `Regional_Dominance`) and applying rigorous Python-based ETL and statistical analysis to over 11,500 historical titles, this project proved that professional critic scores explain only a fraction of commercial success. To maximize ROI in a shrinking physical market, executives must immediately pivot to hyper-targeted regional marketing (e.g., Action in NA, RPGs in JP) and prioritize long-term user community sentiment over chasing superficial critical acclaim.

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16. Appendix

- **Data Dictionary:**
    - `Global_Sales_Calculated`: Sum of NA, EU, JP, Other sales (Validation metric).
    - `Era`: Decadal grouping of `Year_of_Release` (1980s, 1990s, etc.).
    - `Sales_Region_Dominant`: The single region contributing the highest % of sales.
    - `Score_Gap`: `Critic_Score` - (`User_Score` * 10).
- **Python Logic Excerpt (Score Gap):** df["Score_Gap"] = df["Critic_Score"] - (df["User_Score"] * 10).round(0)

- **Raw Cleaning Log:** Complete 10-step ETL log available in `notebooks/02_cleaning.ipynb`

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17. Contribution Matrix (Mandatory)

This section confirms the contribution of each team member across all project stages.

Team Member	*Dataset Sourcing*	*ETL & Cleaning*	*EDA & Analysis*	*Statistical Analysis*	*Tableau Dashboard*	*Report writing*	*PPT & Viva*
Param Khodiyar	✔				✔		✔
Mayank Pillai	✔				✔		✔
Ankita Thakur	✔	✔				✔	✔
Sarabjeet Singh	✔				✔		✔
Anuj Upadhyay	✔			✔			✔
Aditya Yadav	✔		✔		✔		✔

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub commit history and submitted artifacts.

Team Signature Block: Param Khodiyar, Ankita Thakur, Mayank Pillai,Sarabjeet Singh,Anuj Upadhyay,Aditya Yadav